PROPOSED DESIGN / NO NEW MODEL EXECUTION

Select once. Track new frames. Preserve uncertainty.

Phone → host API → isolated learned-model worker. Robot motion stays a separate project.

PHONE / PWA

Existing Preact + TypeScript
Tap / box on pinned frame
Refine actual mask preview
Inspect result age and gaps
Track / Correct / Stop

HOST / API + EVIDENCE

Existing .venv-dev / FastAPI + SQLite
Immutable frame IDs, pixels and camera/session
Ownership + generation + prompt revision
Bounded spool / content-addressed input and mask
No model forward pass in the API

PI / EXISTING RELAY

Camera capture and transport
Robot-side watchdog retained
No new PyTorch model on Pi
No tracking-to-motor command

Commands + input

Mask + frame ID

camera frames

SEARCH / EXISTING MODELS

Separate .venv-vision
Actual YOLOE → DINOv2
New transient-target adapter must still be implemented
Possible match ≠ identity
Automatic recovery gate open

HOST / TRACKING WORKER

Proposed .venv-tracking / explicit CPU or GPU
Original SAM 2.1 image predictor → seed mask
Original Cutie InferenceCore.step → future masks
One processor per session generation
Missing model/device fails; no substitute output
New tracking models: NOT RUN

BOUNDARIES

Old 12-frame jobs are preserved
No old mask on latest camera image
No interpolation to animate a track
No automatic verified room sighting
CPU live latency is not established
No current-GPU or paid-API work

Execution evidence comes before a working-feature claim.

First gate: original SAM + Cutie execution, pinned assets, failure checks, measured quality and latency.

PROPOSED DESIGN / NO NEW MODEL EXECUTION

A session is not proof of identity.

Lifecycle, observation health and human assertions are stored separately.

1 / SELECTING

Freeze the exact frame; bind clicks
Actual SAM mask preview + refinements
Operator chooses the target extent

2 / INITIALIZING

Commit mask revision and generation
Original Cutie + bounded catch-up
Failure stays visible; no substitute

3 / RUNNING

Actual frame + actual mask + timestamp
Tracking · identity unverified
No invented trajectory or confidence

empty mask / health failure / time gap

4 / UNCERTAIN · LOST · STALE

No usable observation under policy
Not proof the target left the room
Health logic is a declared heuristic
Close the affected processor generation

5 / SEARCHING

User requests Find again
Actual YOLOE/DINO candidate evidence
Possible match remains a hypothesis
Automatic recovery requires a new gate

re-prompt

6 / CORRECTION OR RESELECTION

Human input is an intervention
New pinned frame + new generation
Old errors and gaps remain in history
Never count this as automatic recovery

STOPPED / FAILED / INTERRUPTED

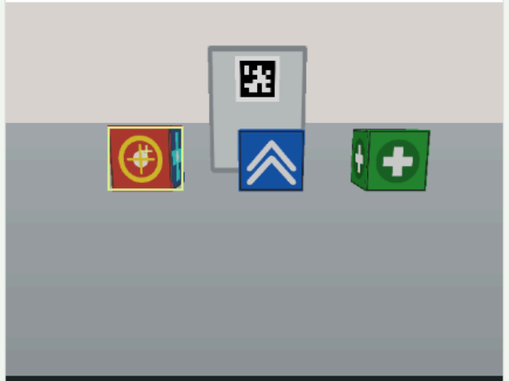
Stop, expiry, worker restart or required-component failure closes publication generation atomically.
A native model call may finish later; its output cannot resurrect the session.
Current location, verified identity, robot following and cross-camera continuity are not inferred from session status.

AUTHORED DESIGN PROPOSAL · NO NEW TRACKING INFERENCE

A target you can point to. A result you can inspect.

Proposed PuppyPi phone workflow: select a target, inspect the model's mask, then track future frames. These screens describe interaction states; they are not a delivered app or model-execution evidence.

PuppyPi
Monitor / target selection
Frozen selection



Synthetic source · authored user box · not a predicted mask

01 / CHOOSE THE TARGET

Is this the area?

Tap the object, draw a box, or exclude the background. The frozen frame keeps your selection in place.

+ Add point | − Exclude | □ Box

Your selected target
Human input · name optional
Model mask: not generated

Preview model mask

Track this object · needs model preview

Cancel selection

Design state only. Actual SAM 2 output is required before Start is enabled.

PuppyPi
Monitor / selected target
Stop tracking

Processed frame + actual mask

Reserved for genuine model output. No result or tracking measurement is invented here.

A mask is shown only on the image that produced it

02 / FOLLOW THE EVIDENCE

Keep the target in view.

The proposed session keeps temporal model memory across new frames. Identity stays unverified.

Observation state	–
Processed capture / age	– / –
Model execution	Not run

Seed ↔ last observation
Two distinct evidence records
Actual observation pending

Correct selection

Stop stays reachable in the top bar.

Design state only. Status, timing and masks must come from a running session.

PuppyPi
Monitor / recovery
Lost state design

No current usable observation

Last evidence belongs in history. No old mask is animated over the live camera.

Loss is not proof of absence from the room

03 / RECOVER WITHOUT GUESSING

Find a possible match.

Search new frames using the selected reference. Inspect candidates and the reasons for uncertainty.

> Re-detection can find similar objects. Automatic identity recovery needs a separate measured acceptance gate.

Loss reason	From session evidence
Candidate search	Not run
Confirmed identity	Not established

Find again

Select again | End session

Design state only. Human reselection counts as an intervention, not automatic recovery.

Implementation boundary: original SAM 2.1 → original Cutie on a separate host worker. Existing YOLOE/DINOv2 serve candidate search. No robot motion is requested. This static board has no working model controls, scores or simulated success. Read the complete design and acceptance plan.